



WP-004

# Technical Validation Framework for Decentralized Organic-Waste Processing

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A release-gated protocol for feedstock qualification, mass balance, process performance, output safety, reliability and pilot evidence

## PUBLIC STATUS

Engineering validation framework. This document does not certify a machine, approve an output, or state verified Rapid Organic performance.

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## Purpose

WP-004 translates Rapid Organic's validation strategy into a public, auditable framework. It defines what must be fixed, measured, sampled, reconciled and reviewed before a technical claim can be released. It is intended for engineers, pilot hosts, laboratories, investors, regulators and commercial partners.

# 1. Executive summary

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A successful demonstration is not the same as a validated product. Technical validation requires a frozen machine configuration, a defined feedstock envelope, calibrated measurements, repeat runs, complete accounting of material and utilities, independent output testing, recorded failures and acceptance criteria agreed before testing.

## Minimum proof package

- Configuration record for hardware, software, controls and operating recipe.
- Representative feedstock characterization and traceable batch identification.
- Closed mass-and-moisture balance with disclosed uncertainty.
- Metered energy, water, additives, labour, cycle time and accepted throughput.
- Accredited laboratory results tied to chain of custody and intended output route.
- Reliability, cleaning, maintenance, alarms, interventions and downtime records.
- Deviation log, raw data archive and signed pass/retest/fail review.

### CORE RULE

Every released claim must identify the tested configuration, feedstock, conditions, method, number of valid runs, result range and limitation.

## 2. Validation principles and claim control

| Principle                     | Required practice                                                                            |
|-------------------------------|----------------------------------------------------------------------------------------------|
| Test the claim                | Convert each proposed claim into a metric, method, boundary and threshold.                   |
| Freeze before measuring       | Do not pool results across undisclosed hardware, software or recipe changes.                 |
| Measure the whole system      | Include preparation, loading, operation, cleaning, discharge and residual routing.           |
| Preserve failures             | Failed and interrupted runs remain part of the evidence record.                              |
| Separate facts from inference | Measured results, calculated results, supplier data and assumptions are labelled distinctly. |
| Release narrowly              | Evidence from one feedstock or model does not automatically apply to another.                |

### Evidence ladder

- Level 0 — design intent or supplier statement.
- Level 1 — observed demonstration with incomplete controls.
- Level 2 — controlled internal repeat testing.
- Level 3 — independent measurements or laboratory verification.
- Level 4 — representative customer pilot under an approved protocol.
- Level 5 — replicated field performance across defined sites and conditions.

#### CLAIM CONTROL

Public language should never exceed the evidence level. “Designed to,” “measured during controlled testing,” and “verified in representative field operation” are different statements.

### 3. Freeze the test article and system boundary

| Configuration item | Record before run release                                                               |
|--------------------|-----------------------------------------------------------------------------------------|
| Mechanical         | Model, serial number, vessel, drive, cutters/mixers, seals, insulation and airflow path |
| Electrical         | Supply, connected load, motors, heaters, protection, meters and panel revision          |
| Controls           | Controller, firmware/software, setpoints, interlocks, alarm logic and data interval     |
| Process recipe     | Load sequence, additives, target states, cycle end rule and discharge method            |
| Site interfaces    | Preparation equipment, ventilation, drainage, storage, utilities and contingency route  |
| Instrumentation    | Tag, range, accuracy, calibration status, location and logging method                   |

#### Change classification

- Editorial: no effect on test validity.
- Minor: documented engineering review confirms no expected effect.
- Major: could affect safety, capacity, balance, output or reliability; retest is required.

#### REFERENCE CONFIGURATION

RO-50 is treated as a validation reference designation until a released specification is supported by completed testing. The name alone is not a capacity guarantee.

## 4. Qualify feedstocks and test conditions

| Domain                | Characterize                                                                     |
|-----------------------|----------------------------------------------------------------------------------|
| Identity              | Source, process origin, segregation practice, age and storage conditions         |
| Physical              | Particle-size distribution, bulk density, free liquid, grit and foreign objects  |
| Composition           | Moisture/total solids, ash, fat, fibre, protein, salt and pH as relevant         |
| Biological / chemical | Known hazards, pathogens, allergens, cleaning chemicals and restricted materials |
| Variability           | Daily, seasonal, supplier and production-line differences                        |
| Acceptance            | Allowed range, rejection criteria, sampling frequency and off-spec route         |

### Test matrix

| Test condition         | Purpose                                                            |
|------------------------|--------------------------------------------------------------------|
| Nominal                | Establish repeatability at the centre of the approved envelope     |
| Boundary low / high    | Test declared moisture, composition or loading limits              |
| Representative mixture | Reflect normal site variation without cherry-picking               |
| Upset challenge        | Evaluate only hazards reviewed and approved in the test plan       |
| Rejection test         | Confirm that prohibited or off-spec inputs are detected and routed |

#### NO UNIVERSAL-FEEDSTOCK CLAIM

Compatibility must be released by feedstock class and tested envelope. One successful batch cannot support “all organic waste.”

## 5. Measurement system and data integrity

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| Measurement                 | Minimum control                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------|
| Mass                        | Traceable scales; tare procedure; capacity and resolution suitable for each stream     |
| Moisture / solids           | Documented sampling, preservation and laboratory or validated field method             |
| Energy                      | Dedicated true-energy meter; include auxiliaries and standby where in boundary         |
| Temperature / process state | Known sensor location, interval, range and calibration check                           |
| Water / additives           | Direct measurement by mass or calibrated volume                                        |
| Time / labour               | Synchronized clock; distinguish cycle, waiting, cleaning, maintenance and intervention |
| Samples                     | Unique ID, chain of custody, retained sample policy and laboratory method              |

### Data integrity controls

- Write a data dictionary with units, tags, frequency and owner.
- Synchronize clocks and retain original machine and manual records.
- Record missing data; never silently interpolate headline results.
- Use pre-defined rules for invalid runs and outliers.
- Lock reviewed datasets and preserve an audit trail for calculations.

## 6. Mass, moisture and material-fate balance

### REQUIRED BALANCE

Inputs = characterized solid output + captured liquids + condensate + estimated exhaust moisture + retained material + samples + cleaning transfers + unexplained difference.

| Ledger               | Record                                                                           |
|----------------------|----------------------------------------------------------------------------------|
| Inputs               | Feedstock, additives, process water, recirculated material and cleaning carry-in |
| Solid outputs        | Product candidate, rejects, retained material and cleaning solids                |
| Liquid outputs       | Free liquid, condensate, wastewater and cleaning liquids                         |
| Atmospheric transfer | Water vapour and any measured or estimated airborne material                     |
| Other                | Samples, spills and documented handling losses                                   |

Report both wet-basis and dry-solids results. A lower wet mass may largely reflect transferred water; it does not by itself prove destruction, stabilization, diversion or value creation.

### Closure and uncertainty

The protocol must set a closure threshold based on scale resolution, sampling uncertainty, moisture variability and estimated streams. Runs outside the threshold are investigated and cannot support material-reduction claims until resolved.



## 7. Process performance and utilities

| Metric              | Expression / reporting rule                                                           |
|---------------------|---------------------------------------------------------------------------------------|
| Accepted throughput | Accepted input kg per scheduled operating day; exclude rejected and bypassed loads    |
| Cycle time          | Start/end definition plus median, range and valid-run count                           |
| Specific energy     | kWh per accepted wet tonne and per input dry-solids tonne                             |
| Peak demand         | Maximum measured demand over declared interval                                        |
| Water / additives   | Quantity per accepted tonne, including cleaning when in boundary                      |
| Material yield      | Characterized output on wet and dry basis                                             |
| Interventions       | Operator actions per run and labour hours per tonne                                   |
| Capacity statement  | Sustained accepted throughput under named feedstock and schedule—not nameplate volume |

### Repeatability

Use multiple valid runs after commissioning and report median, range and variability. Best-run reporting is insufficient. Planned changes require a new test block and configuration record.

#### TWENTY-FOUR-HOUR CAUTION

A cycle completed within 24 hours is only a time result. It does not independently establish output quality, safety, stability, fertilizer status or economic value.

## 8. Output characterization and stewardship

| Question                         | Evidence                                                                                   |
|----------------------------------|--------------------------------------------------------------------------------------------|
| What is it?                      | Composition, moisture/solids, pH, salinity/conductivity, nutrients and physical properties |
| Is it safe?                      | Feedstock-specific biological, chemical and physical hazard testing                        |
| Is it stable?                    | Storage behaviour, re-heating/re-odor, biological activity and shelf/handling observations |
| Is it consistent?                | Between-batch and within-batch sampling across representative operation                    |
| Can it be used lawfully?         | Regulatory classification, claims review and requirements for intended market/use          |
| Where does off-spec material go? | Documented contingency receiver and actual disposition records                             |

### Sampling design

- Define lot and batch before sampling.
- Use a defensible composite or stratified plan where heterogeneity is expected.
- Preserve chain of custody and laboratory certificates.
- Separate screening results from release testing.
- Do not generalize one laboratory result to an undefined production period.

#### TERMINOLOGY DISCIPLINE

Until regulatory status and supporting evidence are established, use “processed output” or “output material,” not fertilizer, compost or certified product.

## 9. Odor, emissions, hygiene and workplace safety

| Domain              | Validation approach                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------|
| Odor                | Assess storage, loading, operation, exhaust, discharge and cleaning; record weather and operating state |
| Air / ventilation   | Review exhaust route, moisture, aerosols, dust and site ventilation requirements                        |
| Hygiene             | Cleaning validation, cross-contamination controls, pests and operator contact points                    |
| Biological exposure | Feedstock hazard assessment, PPE and handling procedures                                                |
| Machine safety      | Guards, access controls, emergency stops, interlocks, lockout and fault response                        |
| Electrical / fire   | Applicable installation review, overload protection, hot surfaces and combustible loading               |
| Wastewater          | Characterize quantity, destination and approval requirements before discharge                           |

Sensory odor logs can support operational comparison but are not a universal “odor-free” proof. Where a permit, complaint risk or claim requires stronger evidence, use an appropriate qualified method and independent reviewer.

# 10. Reliability, maintainability and operability

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| Metric              | Definition                                                          |
|---------------------|---------------------------------------------------------------------|
| Availability        | Run-ready time divided by scheduled required time                   |
| Successful run rate | Runs meeting protocol completion criteria divided by attempted runs |
| Unplanned downtime  | Hours by failure mode and consequence                               |
| Maintenance burden  | Planned and corrective labour, parts, travel and consumables        |
| Cleaning burden     | Time, water, chemicals, waste and access difficulty                 |
| Alarm burden        | Alarm count, cause, duration, operator response and recurrence      |
| Recovery            | Time and steps to return safely to released operation               |
| Operator dependence | Interventions, skill level, training and procedure adherence        |

## Failure discipline

Use consistent reason codes and preserve replaced parts, photographs, logs and corrective actions where practical. Reliability growth is demonstrated by a declining recurrence of defined failure modes—not by excluding difficult days.

# 11. Test phases and acceptance gates

| Phase                           | Primary objective                                                                     | Release gate                              |
|---------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------|
| 0 — Readiness                   | Approve hazards, configuration, methods, feedstock, instrumentation and output routes | Signed test-readiness review              |
| 1 — Commissioning               | Verify installation, controls, interlocks, meters and operator training               | Safe functional acceptance                |
| 2 — Controlled characterization | Establish balances, utilities, repeatability and output profile                       | Engineering review against thresholds     |
| 3 — Envelope testing            | Test approved feedstock and operating boundaries                                      | Released envelope or redesign             |
| 4 — Reliability run             | Operate across planned schedule and representative variation                          | Availability and maintenance evidence     |
| 5 — Customer pilot              | Confirm performance and usability under field conditions                              | Pilot acceptance and limitation statement |
| 6 — Replication                 | Repeat on released build and additional qualified sites                               | Commercial claim or scale decision        |

## PRE-AGREED THRESHOLDS

Acceptance criteria must be approved before results are known. Changing thresholds afterward requires a documented deviation and cannot be hidden as normal interpretation.

# 12. Pilot execution and evidence package

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## Before the pilot

- Signed protocol, responsibility matrix and site baseline.
- Approved feedstock envelope, sample plan and contingency routes.
- Site safety, utilities, drainage, ventilation and permitting review.
- Calibration records, data ownership, confidentiality and publication terms.
- Training, spares, maintenance and stop-work criteria.

## During the pilot

- Retain every attempted run and batch disposition.
- Log alarms, changes, downtime, cleaning and manual intervention.
- Review data completeness daily without rewriting raw records.
- Escalate safety, environmental and acceptance deviations immediately.

## Final evidence package

- Configuration and change history.
- Raw and processed datasets with calculation workbook.
- Laboratory reports and chain-of-custody records.
- Mass/energy balances, performance statistics and uncertainty statement.
- Failure, maintenance, safety and deviation logs.
- Customer observations, actual output disposition and signed decision record.

# 13. Reporting templates and decision rules

| Record            | Required fields                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------|
| Run sheet         | Run ID, configuration, operators, feedstock lot, times, masses, setpoints, utilities and disposition |
| Deviation log     | Requirement, event, impact, immediate control, investigation, owner and closure                      |
| Evidence register | Claim/input, source, date, owner, evidence level, limitation and approved use                        |
| Batch certificate | Identity, sampling, laboratory methods/results, status and destination                               |
| Reliability log   | Failure mode, lost time, part/action, recurrence and verification                                    |
| Decision record   | Criteria, results, unresolved risks, conditions, approvers and next gate                             |

## Decision outcomes

- Pass — all mandatory criteria met within the declared scope.
- Conditional pass — limited release with explicit restrictions and closure actions.
- Retest — evidence invalid or incomplete, but a defined corrective path exists.
- Fail / stop — threshold not met, risk unacceptable, or no lawful route exists.

### NO EVIDENCE LAUNDERING

A conditional pass must not be summarized publicly as an unconditional success.

# 14. Frequently asked questions

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## How many runs are enough?

There is no universal number. The protocol should justify enough valid runs to establish repeatability across the intended variability. More runs do not repair a changing configuration or poor measurements.

## Can supplier data be used?

Yes, as input to planning and risk assessment, if clearly labelled. Rapid Organic release claims require traceable evidence tied to the tested configuration and conditions.

## Does mass reduction prove diversion?

No. Diversion depends on verified final destinations. The balance must show where solids, liquids and moisture went.

## Does laboratory nutrient content make the output fertilizer?

No. Composition is only one part of safety, consistency, regulatory status, claims and intended-use requirements.

## Can an odor claim be made from operator observation?

Only a narrow observation under the recorded conditions. A broad “odor-free” claim requires a defined, appropriate evidence method and scope.

## When must testing be repeated?

After a major design, software, recipe or feedstock change; after unresolved measurement failure; or when expanding beyond the released envelope.

## What is the most important first test?

A controlled set of repeat runs on one frozen RO-50 reference configuration and one representative, well-characterized feedstock, with complete balance and output sampling.

## Who decides pass or fail?

The approved protocol assigns technical, safety, laboratory, site and commercial authorities. The evidence review should include the pilot host and independent expertise where the claim or risk requires it.



# 15. Conclusion

Rapid Organic's strongest path to credibility is disciplined, narrow proof. Freeze one configuration. Qualify one feedstock. Measure the entire system. Reconcile material and moisture. Test outputs independently. Preserve failures. Release only the claims that the evidence supports. Then repeat under representative customer conditions before scaling.

## NEXT VALIDATION MILESTONE

Complete controlled repeat testing of the frozen RO-50 reference configuration, establish an output characterization dossier, and proceed to a Canadian flagship pilot only after the technical release gate is met.

## Selected public references

- [ISO/IEC 17025 — Competence of testing and calibration laboratories](#)
- [ISO 5725 — Accuracy of measurement methods and results](#)
- [CFIA — Fertilizer and supplement guidance](#)
- [CCME — Guidelines for Compost Quality](#)
- [CCOHS — Hazard control and workplace safety resources](#)
- [Rapid Organic — Public Knowledge Base](#)

## Document control

| Field            | Value                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------|
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